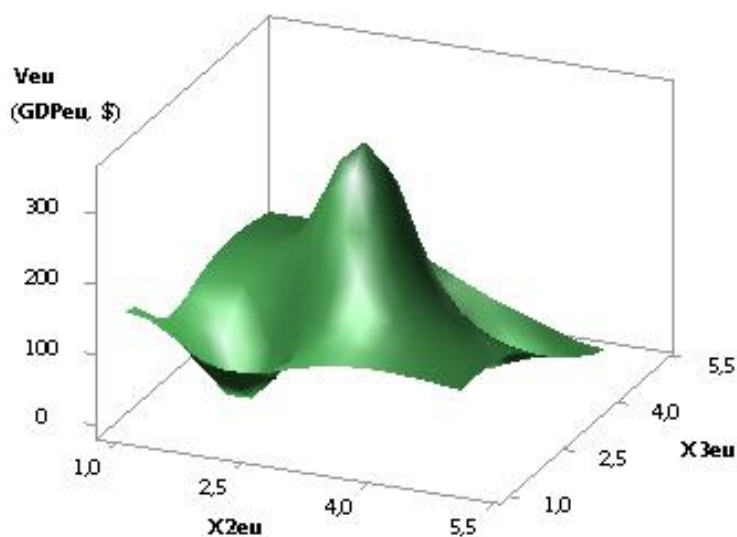


Э. А. ПИЛЬ

**ТЕОРИЯ
ЭКОНОМИЧЕСКИХ КРИЗИСОВ
THE THEORY OF ECONOMIC
CRISES**



**САНКТ-ПЕТЕРБУРГ
2017**

'For me a word of wisdom is more valuable than any gold.'

Democritus

INTRODUCTION

Various authors dedicated a huge amount of their works to the issue of economic and financial crisis and offered their ways out of the crisis. Although, unfortunately, none of works succeeded in the profound solving of the problem. Due to this fact the author of this work suggests to use the theory of the economic shell in order to describe processes in economy suffering from crisis. This theory gave an opportunity to design theoretic scenarios of economic crisis development and to perform calculations which help to indicate the probable methods to be used in recovering from any type of economic recession. These methods enable government to choose an appropriate way out of the particular crisis without loss and even with the increase of gross domestic product of a country.

CONTENTS

INTRODUCTION

CHAPTER 1

THE DEVELOPMENT OF ECONOMIC CRISES AND THEIR ANALYSIS

1. Classification of Economic Crises	10
1.1 Classification of Economic Crises – 1	11
By scale of equilibrium disruption in economic systems	11
By regularity of equilibrium disruption in the economy	11
By the nature of reproduction proportion violations	12
1.2. Classification of Economic Crises – 2	12
By causes of occurrence	12
By duration	12
By stages	12
By sphere of occurrence	12
By nature	12
By scale	13
By depth	13
By periodicity	13
1.3. Classification of Economic Crises – 3	13
By scale of manifestation	13
By crisis issues	13
By structure of relations in the socio-economic system	13
By immediate causes of occurrence	14
By nature of manifestation	14
By time factor	14
1.4. Classification (Typology) of Crises – 4	14
Crises	14
Functions of a crisis	15
Crisis dynamics	15
Organizational crises	16
Crisis factors in an organization	17
Grouping of factors of the organization's external and internal environment	17
Types, phases, and cycles of crisis in an organization	17
Organization's obligations and its insolvency as main causes of crisis	18
Public administration crises	19
Features of economic crisis at the macroeconomic level	19
1.5. Classification of Crises – 5	20
1.6. Typology of Crises	20
2. Analysis of Theories of Economic Crises	23
2.1. Formation of Crisis Theory	23
Evolution of classical crisis theory	26
Economic crises from the perspective of financial intermediation theory	30
Crises and economic survival	31
2.2. Unified Theory of Cycles, Crises, and Formations	34
2.3. Comparative Analysis of Theories of the Origin of the Modern Financial-Economic Crisis	34
by M. Khazin and N. Taleb	34

CONCLUSION

CHAPTER 2

THEORETICAL POSSIBILITIES OF CHANGES IN ECONOMY AND POPULATION

2.1. Construction of a Table of Variants for Economic and Population Development	40
2.2. Variants of Development of Variables <i>et</i> and <i>pt</i>	45
2.3. Classification of Development of Variables <i>et</i> and <i>pt</i>	69
2.4. Classification of Countries by Areas Occupied by Economy and Population	69
CONCLUSION	73

CHAPTER 3

VARIANTS OF MACROECONOMICS DEVELOPMENT AFTER BEING AFFECTED BY EXTERNAL AND INTERNAL PRESSURES

3.1 Variants of Macroeconomics Development After Being Affected by External and Internal Pressures	74
3.2. Spherical Economic Shells	88
3.3. Ellipsoidal Economic Shells	91
3.4. Variants for Calculating the Volume of a Spherical Economic Shell	91
3.5. Variants for Calculating the Surface Area of a Spherical Economic Shell	95
CONCLUSION	101

CHAPTER 4

CALCULATION OF PARAMETERS OF VOLUME OF A SPHERICAL ECONOMIC SHELL

4.1. Upper Critical Volume V_{su} of a Spherical Economic Shell	102
4.1.1. Calculation of the Upper Critical Volume V_{su} of a Spherical Economic Shell	102
4.1.2. Summary Tables of the Upper Critical Volume V_{su} of a Spherical Economic Shell	152
4.1.3. Calculation of the Upper Critical Volume V_{su} with Significant Decrease in All Variable Values	158
4.1.4. Calculation of the Upper Critical Volume V_{su} with Increase in the Number of Decimal Places in Variable X_{4su}	160
4.1.5. Calculation of the Unit Theoretical Volume of the Upper Critical Value V_{stu1} of a Spherical Economic Shell	161
4.2. Lower Critical Volume V_{sl} of a Spherical Economic Shell	164
4.2.1. Calculation of the Lower Critical Volume V_{sl} of a Spherical Economic Shell	164
4.2.2. Summary Tables of the Lower Critical Volume V_{sl} of a Spherical Economic Shell	177
4.2.3. Calculation of the Lower Critical Volume V_{sl} with Significant Decrease in All Variable Values	181
4.2.4. Calculation of the Unit Theoretical Volume of the Lower Critical Value V_{stl1} of a Spherical Economic Shell	182
4.3. Existence Area of Volume ΔV_{sul} of a Spherical Economic Shell	185
4.3.1. Calculation of the Existence Area of Volume ΔV_{sul} of a Spherical Economic Shell	185
4.3.2. Summary Tables of the Existence Area of Volume ΔV_{sul} of a Spherical Economic Shell	198
4.4. Upper Critical Existence Area of Volume SV_{su} of a Spherical Economic Shell	202
4.4.1. Calculation of the Upper Critical Existence Area of Volume SV_{su} of a Spherical Economic Shell	202
4.4.2. Summary Tables of the Upper Critical Existence Area of Volume SV_{su} of a Spherical Economic Shell	229
4.5. Lower Critical Existence Area of Volume SV_{sl} of a Spherical Economic Shell	235
4.5.1. Calculation of the Lower Critical Existence Area of Volume SV_{sl} of a Spherical Economic Shell	235
4.5.2. Summary Tables of the Lower Critical Existence Area of Volume SV_{sl} of a Spherical Economic Shell	244
4.6. Existence Area of Volume ΔV_{sul} of a Spherical Economic Shell	247
4.6.1. Calculation of the Existence Area of Volume ΔV_{sul} of a Spherical Economic Shell	247
4.6.2. Summary Tables of the Existence Area of Volume ΔV_{sul} of a Spherical Economic Shell	256

CHAPTER 5

CALCULATION OF PARAMETERS OF SURFACE AREA OF A SPHERICAL ECONOMIC SHELL

5.1. Upper Critical Surface Area S_{su} of a Spherical Economic Shell	259
5.1.1. Calculation of the Upper Critical Surface Area S_{su} of a Spherical Economic Shell	259
5.1.2. Summary Tables of the Upper Critical Surface Area S_{su} of a Spherical Economic Shell	287
5.1.3. Calculation of the Upper Critical Surface Area S_{su} with Significant Decrease in All Variable Values	294
5.1.4. Calculation of the Upper Critical Surface Area S_{su} with Increase in the Number of Decimal Places in Variable X_{4su}	295
5.1.5. Calculation of the Unit Theoretical Upper Critical Surface Area S_{stu1} of a Spherical Economic Shell	296
5.2. Lower Critical Surface Area S_{sl} of a Spherical Economic Shell	299
5.2.1. Calculation of the Lower Critical Surface Area S_{sl} of a Spherical Economic Shell	299
5.2.2. Summary Tables of the Lower Critical Surface Area S_{sl} of a Spherical Economic Shell	309
5.2.3. Calculation of the Surface Area S_{sl} with Significant Decrease in All Variable Values	312
5.2.4. Calculation of the Unit Theoretical Lower Critical Surface Area S_{stl1} of a Spherical Economic Shell	313
5.3. Existence Area of Surface Area ΔS_{sul} of a Spherical Economic Shell	317

5.3.1. Calculation of the Existence Area of Surface Area ΔS_{su} of a Spherical Economic Shell	317
5.3.2. Summary Tables of the Existence Area of Surface Area ΔS_{su} of a Spherical Economic Shell	326
5.4. Upper Critical Existence Area of Surface Area S_{su} of a Spherical Economic Shell	330
5.4.1. Calculation of the Upper Critical Existence Area of Surface Area S_{su} of a Spherical Economic Shell	330
5.4.2. Summary Tables of the Upper Critical Existence Area of Surface Area S_{su} of a Spherical Economic Shell	357
5.5. Lower Critical Existence Area of Surface Area S_{sl} of a Spherical Economic Shell	363
5.5.1. Calculation of the Lower Critical Existence Area of Surface Area S_{sl} of a Spherical Economic Shell	363
5.5.2. Summary Tables of the Lower Critical Existence Area of Surface Area S_{sl} of a Spherical Economic Shell	372
5.6. Existence Area of Surface Area ΔS_{su} of a Spherical Economic Shell	375
5.6.1. Calculation of the Existence Area of Surface Area ΔS_{su} of a Spherical Economic Shell	375
5.6.2. Summary Tables of the Existence Area of Surface Area ΔS_{su} of a Spherical Economic Shell	384
5.7. Comparison of Parameter Values of Volume V_{su} and Surface Area S_{su}	387
5.7.1. Calculation of Comparisons of Parameter Values of Volume V_{su} and Surface Area S_{su}	387
5.7.2. Summary Tables of Volume V_{su} and Surface Area S_{su} of a Spherical Economic Shell	405
5.8. Comparison of Parameter Values of Volume V_{sl} and Surface Area S_{sl}	409
5.8.1. Calculation of Comparisons of Parameter Values of Volume V_{sl} and Surface Area S_{sl}	409
5.8.2. Summary Tables of Volume V_{sl} and Surface Area S_{sl} of a Spherical Economic Shell	413
CHAPTER 6	
CALCULATION OF PARAMETERS OF VOLUME OF AN ELLIPSOIDAL ECONOMIC SHELL	414
6.1. Upper Critical Volume V_{eu} of an Ellipsoidal Economic Shell	414
6.1.1. Calculation of the Upper Critical Volume V_{eu} of an Ellipsoidal Economic Shell	414
6.1.2. Summary Tables of the Upper Critical Volume V_{eu} of an Ellipsoidal Economic Shell	491
6.1.3. Calculation of the Upper Critical Volume V_{eu} with Significant Decrease in All Variable Values	498
6.1.4. Calculation of the Upper Critical Volume V_{eu} with Increase in the Number of Decimal Places in Variables X_{4eu} and X_{6eu}	500
6.1.5. Calculation of the Unit Theoretical Volume of the Upper Critical Value V_{eu1} of an Ellipsoidal Economic Shell	504
6.2. Lower Critical Volume V_{el} of an Ellipsoidal Economic Shell	507
6.2.1. Calculation of the Lower Critical Volume V_{el} of an Ellipsoidal Economic Shell	507
6.2.2. Summary Tables of the Lower Critical Volume V_{el} of an Ellipsoidal Economic Shell	563
6.2.3. Calculation of the Lower Critical Volume V_{el} with Significant Decrease in All Variable Values	569
6.2.4. Calculation of the Lower Critical Volume V_{el} with Increase in the Number of Decimal Places in Variable X_{5el}	570
6.2.5. Calculation of the Unit Theoretical Volume of the Lower Critical Value V_{el1} of an Ellipsoidal Economic Shell	571
6.3. Existence Area of Volume ΔV_{eu} of an Ellipsoidal Economic Shell	574
6.3.1. Calculation of the Existence Area of Volume ΔV_{eu} of an Ellipsoidal Economic Shell	574
6.3.2. Summary Tables of the Existence Area of Volume ΔV_{eu} of an Ellipsoidal Economic Shell	606
6.4. Upper Critical Existence Area of Volume S_{Veu} of an Ellipsoidal Economic Shell	612
6.4.1. Calculation of the Upper Critical Existence Area of Volume S_{Veu} of an Ellipsoidal Economic Shell	612
6.4.2. Summary Tables of the Upper Critical Existence Area of Volume S_{Veu} of an Ellipsoidal Economic Shell	661
6.5. Lower Critical Existence Area of Volume S_{Vel} of an Ellipsoidal Economic Shell	667
6.5.1. Calculation of the Lower Critical Existence Area of Volume S_{Vel} of an Ellipsoidal Economic Shell	667
6.5.2. Summary Tables of the Lower Critical Existence Area of Volume S_{Vel} of an Ellipsoidal Economic Shell	706
6.6. Existence Area of Volume ΔV_{eu} of an Ellipsoidal Economic Shell	711
6.6.1. Calculation of the Existence Area of Volume ΔV_{eu} of an Ellipsoidal Economic Shell	711
6.6.2. Summary Tables of the Existence Area of Volume ΔV_{eu} of an Ellipsoidal Economic Shell	749

CHAPTER 7

CALCULATION OF PARAMETERS OF SURFACE AREA OF AN ELLIPSOIDAL ECONOMIC SHELL

	755
7.1. Upper Critical Surface Area S_{eu} of an Ellipsoidal Economic Shell	755
7.1.1. Calculation of the Upper Critical Surface Area S_{eu} of an Ellipsoidal Economic Shell	755
7.1.2. Summary Tables of the Upper Critical Surface Area S_{eu} of an Ellipsoidal Economic Shell	797
7.1.3. Calculation of the Upper Critical Surface Area S_{eu} with Significant Decrease in All Variable Values	804
7.1.4. Calculation of the Upper Critical Surface Area S_{eu} with Increase in the Number of Decimal Places in Variables X_{4eu} and X_{6eu}	806
7.1.5. Calculation of the Unit Theoretical Surface Area of the Upper Critical Value $Setu1$ of an Ellipsoidal Economic Shell	808
7.2. Lower Critical Surface Area S_{el} of an Ellipsoidal Economic Shell	810
7.2.1. Calculation of the Lower Critical Surface Area S_{el} of an Ellipsoidal Economic Shell	810
7.2.2. Summary Tables of the Lower Critical Surface Area S_{el} of an Ellipsoidal Economic Shell	842
7.2.3. Calculation of the Lower Critical Surface Area S_{el} with Significant Decrease in All Variable Values	848
7.2.4. Calculation of the Surface Area S_{el} with Increase in the Number of Decimal Places in Variable X_{5el}	849
7.2.5. Calculation of the Unit Theoretical Lower Critical Surface Area $Setl1$ of an Ellipsoidal Economic Shell	850
7.3. Existence Area of Surface Area ΔS_{eu} of an Ellipsoidal Economic Shell	852
7.3.1. Calculation of the Existence Area of Surface Area ΔS_{eu} of an Ellipsoidal Economic Shell	852
7.3.2. Summary Tables of the Existence Area of Surface Area ΔS_{eu} of an Ellipsoidal Economic Shell	884
7.4. Upper Critical Existence Area of Surface Area S_{Seu} of an Ellipsoidal Economic Shell	890
7.4.1. Calculation of the Upper Critical Existence Area of Surface Area S_{Seu} of an Ellipsoidal Economic Shell	890
7.4.2. Summary Tables of the Upper Critical Existence Area of Surface Area S_{Seu} of an Ellipsoidal Economic Shell	939
7.5. Lower Critical Existence Area of Surface Area S_{Sel} of an Ellipsoidal Economic Shell	946
7.5.1. Calculation of the Lower Critical Existence Area of Surface Area S_{Sel} of an Ellipsoidal Economic Shell	946
7.5.2. Summary Tables of the Lower Critical Existence Area of Surface Area S_{Sel} of an Ellipsoidal Economic Shell	984
7.6. Existence Area of Surface Area $S_{\Delta S_{eu}}$ of an Ellipsoidal Economic Shell	989
7.6.1. Calculation of the Existence Area of Surface Area $S_{\Delta S_{eu}}$ of an Ellipsoidal Economic Shell	989
7.6.2. Summary Tables of the Existence Area of Surface Area $S_{\Delta S_{eu}}$ of an Ellipsoidal Economic Shell	1027
CONCLUSION	1033
LIST OF ABBREVIATIONS	1034
REFERENCES	1040
ADDENDUM	1049
BIOGRAPHY	1064